

Evaluation of Collaborative Learning Effectiveness and Role Identification in Multimodal Data Fusion

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Abstract—In collaborative learning, multimodal data fusion has received extensive interest because it has the benefit of incorporating the heterogeneous information of multiple sources. In this paper, a multimodal collaborative learning modeling approach will be proposed anchored on the feature selection and attention processes. The dimensionality of the initial multimodal data is first reduced by principal component analysis (PCA) and the redundant information is removed. After that, it is followed by a multi-head self-attention-based mechanism which facilitates the fusion of cross-modal features, promoting the representational capacity of key features. Then, a fused feature-based collaborative learning network that is trained end-to-end using role labels is built and trained to identify and classify collaborative roles automatically. The experimental outcomes on the CMU-MOSEI dataset and an own-constructed collaborative dataset demonstrate that the proposed approach upraises the weight of the behavioral log features up to 32.5 percent in the multimodal feature contribution analysis. The error of misclassifying between organizers and collaborators in the role recognition confusion matrix is reduced to 4.1%; the recall rate of the dynamic role transfer identification in the collaboration is 84.3%, which means that fine-grained processes and dynamic changes of functions in collaborative learning have higher perception and representation capacity.

Keywords—multimodal data fusion, collaborative learning, role recognition, feature selection, attention mechanism.

I. INTRODUCTION

Collaborative learning as a significant agent of collective intelligence formation has slowly changed its research objectives of macro behavioral patterns to the micro processes of interactions. The introduction of multimodal data also increases the description of the learning process based on both text and behavioral records to multi-source information dimensions including speech, images, expressions, and operation trajectories and, therefore, demonstrates cognitive variations and collaborative aspects of learners at a more realistic and fine-grained level. Nevertheless, the heterogeneity of multimodal information and the noise in this information represent an intrinsic bottleneck of the collaborative learning analysis research to derive effective features to comprehend role relations and dynamical variations of high-dimensional and multi-source information.

The paper concentrates on multimodal information fusion and develops a collaborative learning model modeled by fusion aspects, specifically on how the fusion issues of multisource features are solved concerning the semantic alignment issues and weight distribution problems. A complementary relationship between modalities can be reinforced by introducing a fusion strategy based on the selection of features and attention mechanism, eliminating the redundant features and accurate characterization of learning role behavior patterns can be obtained. In terms of the method

of study, the given research does not only investigate the possible usefulness of multimodal data in complicated collaborative settings, but it also offers novel technological avenues and concepts of intelligent learning conduct identification and procedure tracking.

The general structure of this article is the following: Section 1 expounds the research background, formulation of the problem and its importance in the studies; Section 2 reviews the relevant works and summarizes the research developments in the sphere of multimodal learning analysis and role modeling; Section 3 elaborates the proposed methods including dimensionality reduction of features, mechanism of attention fusion, and process of model construction; Section 4 presents the experimental design and result analysis, compares the performances of the models under different algorithms, and discusses it with the help of confusion matrix and degree of contribution of the features; Section 5 summarizes the research results, proposes improvement directions and future research prospects.

II. RELATED WORK

Collaborative learning, as an important pathway for group cognition and knowledge construction, involves multiple variable interactions and complex path mechanisms. Different academic achievements have revealed the influencing factors and mechanisms in the collaborative process from both macro and micro perspectives, forming a multi-angle theoretical and empirical exploration. Zhu and Luo [1] used the fuzzy set qualitative comparative analysis method to explore the multiple interaction effects and influence paths of internal and external factors on the effectiveness of collaborative learning from a systematic perspective. The following results were obtained: social interaction, teacher support, self-efficacy, learning resources, and learning motivation have a positive impact on the effectiveness of collaborative learning under the dual line teaching mode, and different groups of the five factors play different roles under different paths.

He [2] revealed from a micro perspective the multiple concurrent paths through which the evolution of generative roles affected the effectiveness of knowledge point learning, and proposes four configurations in which the evolution of generative roles affected the effectiveness of knowledge point learning: content interaction, organizational instruction, composite influence, and media influence. Wu et al. [3] collected all online collaborative session data of 31 learners in a computational thinking course at a certain normal university, and used qualitative comparative analysis to explore the conditional configuration that affects learning outcomes. Based on research data, Wei and Yang [4] used the "Statistics" course of the International Economics and Trade major in HB University for the 2019-2020 cohort as a case study. Starting from the dimensions of "teaching methods" and "self-directed

learning", they used the fuzzy set qualitative comparative analysis method (fsQCA) to explore the transmission path that affects the learning effectiveness of college students' courses. Huang [5] summarized a task assignment strategy suitable for VR collaborative learning, developed a VR multiplayer collaborative education system, and applied this strategy to automatically assign tasks. Costley [6] aims to expand existing research on the relationship between collaboration and related cognitive load in online learning environments. Yang [7] briefly reviews the history of collaborative learning and cooperative learning to clarify their origins, differences, and similarities.

Warsaw et al. [8] aimed to investigate the impact of cooperative learning on learners' critical thinking skills and their ability to maintain critical thinking skills in response to Islamic extremism. Sujatha and Vinayakan [9] explored the role of cooperative learning in mathematics and evaluated its impact on learning outcomes by drawing on a series of theoretical frameworks and empirical research. Xu et al. [10] conducted a meta-analysis of 36 relevant literature published in global education journals in the 21st century, aiming to explore the effectiveness of collaborative problem-solving in promoting students' critical thinking, and based on existing evidence, determine to what extent collaborative problem-solving leads to an increase or decrease in critical thinking. Although existing achievements have revealed the multi-source influence structure of collaborative learning effectiveness, the focus on static relationships and local paths is still insufficient for comprehensive modeling of dynamic role transfer, modal interaction, and temporal features. Theoretical deepening and model expansion are worthy of further strengthening.

III. METHOD

A. Multimodal Data Acquisition and Preprocessing

The collection of multimodal data covers multi-source heterogeneous information in collaborative learning environments, including conversation texts, screen images, and interaction behavior logs. The semantic vector representation of text data is obtained through word segmentation, stop word filtering, and word embedding mapping, and the semantic content is transformed into a numerical matrix using a word vector model.

The image modality corresponds to collaborative interface screenshots and facial expression frames, which are pre-trained by ResNet network to extract deep features, and then constructed into temporal visual representations through mean pooling. The behavior log is recorded in real-time by the learning platform, including click streams, window switching durations, and operation sequences. The behavior sequence is sliced using a sliding window to generate behavior embeddings. The feature dimensions obtained from different modalities differ significantly, and direct fusion can lead to uneven information and overfitting. Therefore, principal component analysis is used to reduce the dimensionality of each modality feature. For any modal feature matrix as shown in Equation (1):

$$X=[x_1, x_2, \dots, x_n]^T \in R^{n \times d} \quad (1)$$

By calculating the covariance matrix $C = \frac{1}{n-1} (X - \bar{X})^T (X - \bar{X})$, the eigenvalue decomposition $C = PAP^T$ is obtained. The top k

principal components with variance contribution rates greater than 95% are retained to obtain the reduced dimensional representation as shown in Equation (2):

$$Z = XP_k \quad (2)$$

Where P_k is the orthogonal basis of the corresponding principal component. This process reduces non discriminative noise and reduces inter modal feature redundancy while maintaining key discriminative information [11]. By aligning the reduced dimensional text, image, and behavior vector space, consistency in statistical distribution of different information sources can be achieved

B. Multimodal Feature Fusion

Multimodal feature fusion relies on a multi-head self attention mechanism to capture the interactive dependencies between different information sources. After aligning the dimensions of text, image, and behavior features, they are concatenated into a unified representation sequence and input into the multi-head attention module. Each attention head independently calculates the query matrix Q_i , key matrix K_i , and value matrix V_i , and dynamically assigns fusion weights based on feature similarity. For the i -th attention head, attention mapping is defined as shown in Equation (3).

$$Attention(Q_i, K_i, V_i) = Softmax(\frac{Q_i K_i^T}{\sqrt{d_k}}) V_i \quad (3)$$

Where d_k is a feature dimension scaling parameter used to prevent gradients from being too large. Multi-head attention enhances feature sensitivity to different modal subspaces through parallel computing, and finally outputs a fusion matrix after concatenation and linear transformation as shown in Equation (4).

$$H = [h_1; h_2; \dots; h_m] W_o \quad (4)$$

Where h_m is the output of the m th head, and W_o represents the projection matrix. The fusion result enhances stability through layer normalization and residual mapping, making semantic associations and behavioral dynamics distinguishable in the same representation space [12-13].

In actual training, the attention weights of various modal features show a differential distribution. Text features have higher weights in semantic scene recognition, while behavioral features have an increased proportion in collaborative mode judgment. Table 1 shows the average feature weight distribution of the multi-head attention mechanism under six attention heads.

TABLE I. MEAN DISTRIBUTION OF FEATURE WEIGHTS

Attention Head	Text Modality	Image Modality	Behavioral Modality	Mean Variance
Head 1	0.42	0.27	0.31	0.012
Head 2	0.38	0.33	0.29	0.009
Head 3	0.45	0.24	0.31	0.015
Head 4	0.40	0.30	0.30	0.010
Head 5	0.37	0.29	0.34	0.011
Head 6	0.39	0.25	0.36	0.014

C. Collaborative Learning Modeling Driven by Fusion Features

Based on end-to-end neural networks, automatic recognition and dynamic updating of collaborative roles are

achieved through structured fusion representation. The fusion feature matrix is jointly encoded by multiple layers of perception units and temporal modeling units, with the front-end consisting of two layers of nonlinear transformations to enhance feature discrimination. The input feature vector is denoted as $H \in \mathbb{R}^{n \times d}$, and is processed through linear mapping and activation function to obtain as shown in Equation (5):

$$F = \sigma(W_1 H + b_1) \quad (5)$$

Where $\sigma(\cdot)$ is the GELU activation function. The output of the encoding layer is passed through a bidirectional gated recurrent network to capture the temporal correlation during the collaboration process, and the update state is defined as shown in Equation (6).

$$h_t = \text{BiGRU}(F, h_{t-1}) \quad (6)$$

Where h_t represents the dynamic representation of the character at time step t . The tail of the network adopts multiple classification output units to achieve role probability

prediction, and the classifier calculates the distribution of role labels through the Softmax function as shown in Equation (7):

$$p(y_t / h_t) = \text{Softmax}(W_2 h_t + b_2) \quad (7)$$

Where $y_t \in \{r_1, r_2, r_3\}$ correspond to the role labels of organizer, collaborator, and observer, respectively. The training process is based on minimizing cross entropy loss and dynamically adjusting the learning rate with Adam optimizer to achieve stable convergence.

To address the uneven distribution of roles, learnable weights are introduced into the loss function to balance sample contributions [14-15]. The model parameters are jointly optimized through multimodal fusion of feature inputs, gradually forming a nonlinear representation of role behavior patterns in collaborative learning contexts, enabling the system to have adaptive recognition capabilities for differences in organizational structure, task allocation, and interaction intensity. Figure 1 shows the modeling process of this article:

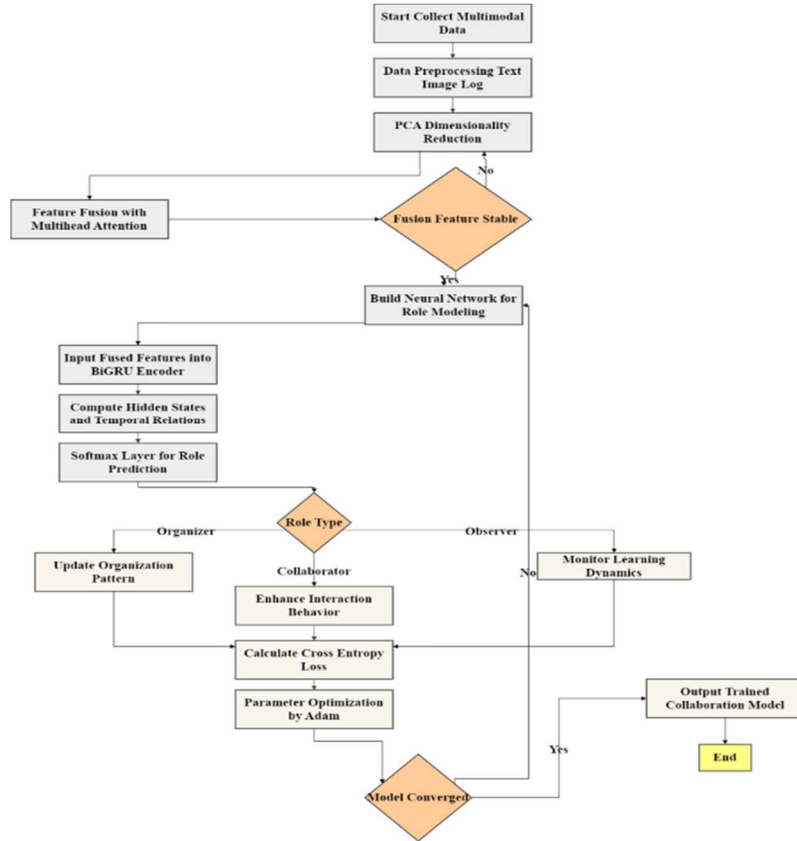


Fig. 1. Modeling Process Diagram

D. Automatic Role Recognition and Dynamic Migration Detection Methods

The hidden state sequence encoded by BiGRU is mapped to the character probability space, and the collaborative behavior pattern is implicitly modeled using time-dependent context. The system calculates the differences in character distribution between adjacent time slices through a sliding window strategy to capture potential migration trends in character identity. For the role representation vector r_t of any learner in the time interval $t, t + \Delta t$, the degree of difference is quantified in Euclidean distance form as shown in Equation (8):

$$D_t = \|p(y_{t+\Delta t}) - p(y_t)\|_2 \quad (8)$$

When D_t exceeds the dynamic threshold θ , the model determines that the character state has changed. The threshold adopts an adaptive update mechanism, which determines the balance distribution between different roles through statistical analysis of the training set, in order to cope with the differences in transfer sensitivity under different task complexities. The dynamic detection part shares parameters with the recognition network, and relies on time attention pooling to further enhance the effectiveness of long-range dependency learning.

IV. RESULTS AND DISCUSSION

A. Introduction to Experimental Dataset

The experiment uses the publicly available multimodal emotion understanding dataset CMU-MOSEI and a self built collaborative learning dataset as the main validation sources. CMU-MOSEI contains 23453 video samples, covering language, visual, and audio modalities. The sampling objects are online courses and interview clips, used to verify the effectiveness of the model's feature fusion in heterogeneous multimodal scenarios. The self built dataset comes from the collaborative project process of the blended learning platform in universities, which includes interaction data of 96 groups

and 288 learners. The recording period is 8 weeks, and the content includes text discussion records, interface screenshot images, and system behavior logs. The experimental environment is deployed on Ubuntu 22.04 servers equipped with NVIDIA RTX 3090 GPUs, featuring Inter i9-13900K CPUs and 128GB of memory. All models are implemented using PyTorch 2.1 with Adam optimizer, batch size set to 32, and maximum training epochs set to 100. The experimental scenario covers two conditions: real classroom and online collaborative tasks, ensuring that the data distribution is both natural and representative.

B. Multimodal Feature Contribution Analysis

TABLE II. CONTRIBUTION ANALYSIS RESULTS

Modality	Dimension Mean	Weight Range	Variance	Mean Contribution(%)
Text Feature	0.58	0.43–0.67	0.012	31.4
Image Feature	0.49	0.35–0.61	0.015	26.1
Behavior Log	0.63	0.52–0.74	0.010	32.5
Audio Auxiliary	0.41	0.29–0.53	0.018	22.8
Fusion Residual Term	0.33	0.22–0.46	0.020	17.2
Global Average	0.49	0.36–0.60	0.015	26.0

The contribution of multimodal features is calculated by integrating attention weights and normalizing feature importance. After model training, the attention matrices of each modality are extracted, and the gradient weighted class response mapping (Grad CAM) and Shapley value are used to jointly estimate the proportion of feature influence on the final classification result. Then, the different modal components are normalized to obtain the average contribution percentage value, in order to evaluate their information gain in the angular color recognition task. Table 2 shows the results of contribution analysis:

Through experimental analysis, the log features of behavior make the greatest contribution of 32.5% to fusion learning, indicating that there is a high sensitivity to variations in interaction timing. Text features are next and have 31.4%, but they do not fluctuate in semantic representation. The context joint modeling is more essential to support the image and audio modalities and the input of the single-mode is quite low. As much as the contribution of the fusion residual term is low, it can enhance the balance of the model and.

C. Role Recognition Confusion Matrix and Misjudgment Rate

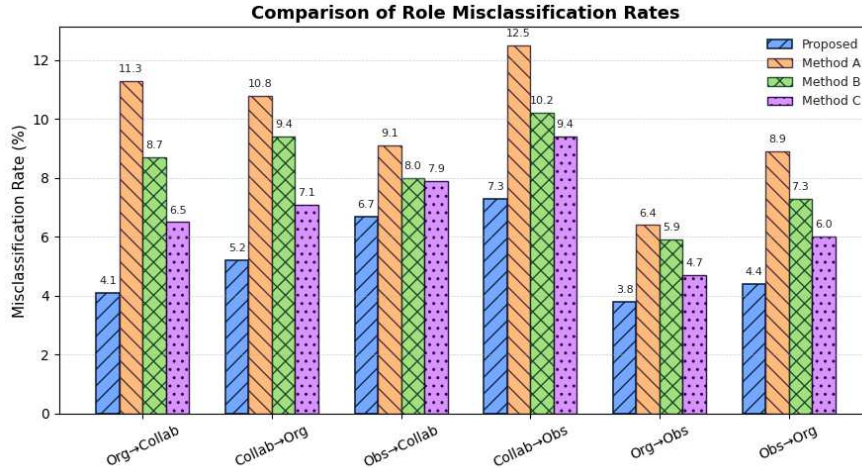


Fig. 2. Misjudgment rate results

The confusion matrix results of role recognition show that the fused feature model has high discrimination between different roles, especially with a significantly reduced misjudgment rate of 4.1% between organizers and collaborators. To verify the effectiveness of the method, a comparison was made between the traditional feature stitching model (Method A), the unimodal attention model (Method B), and the multimodal alignment network (Method C). Figure 2 shows the results of the misjudgment rate analysis:

The analysis results show that the model in this article maintains the lowest level of misjudgment in all types of conversions, with a misjudgment rate of 4.1% between

organizers and collaborators and an average misjudgment rate of 5.25%, which is 4.58 percentage points lower than Method A. The fusion of behavior log features and multi-head attention effectively suppresses role confusion caused by semantic ambiguity. Method B has a tendency towards overfitting in semantic discrimination, and there is still a high degree of misjudgment between organizers and collaborators (8.7%). Method C performs well in structural feature modeling, but lacks the ability to perceive temporal dynamics. The overall trend indicates that the fusion strategy proposed in this article can significantly improve the stability and accuracy of character boundary recognition.

D. Dynamic Role Migration Detection Performance

In the dynamic role transfer detection experiment, the model parameters are frozen and the fusion feature cache is constructed before testing. The same training epochs and learning rate are uniformly used to ensure comparability between different models. The test set selected a randomized collaborative scene sequence with a total of 1520 sample segments and a time window of 12 frames to evaluate the stability and sensitivity of character transformation detection. Figure 3 shows the recall test data for dynamic role migration detection:

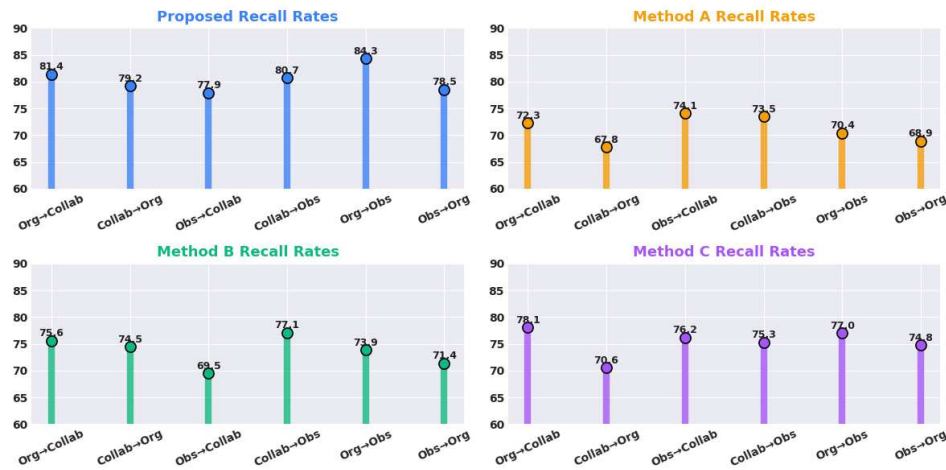


Fig. 3. Recall rate test data for dynamic role migration detection (%)

V. CONCLUSION

The multimodal fusion modeling method proposed in this article effectively alleviates the redundancy and conflict problems of heterogeneous features in the representation layer of collaborative learning. Through feature selection and attention mechanisms, deep interaction and efficient alignment of semantic, behavioral, and visual information are achieved, enabling the model to accurately capture dynamic relationship changes between roles in complex collaborative contexts, demonstrating stronger situational sensitivity and structural interpretability. This method balances the risk of information dilution and overfitting in feature fusion strategy, making the model learning process more stable and able to more finely characterize the role transfer patterns and interactive feature differences of learners in group collaboration. However, there are still issues of sensitivity to noise samples and insufficient modeling ability for long-term temporal dependencies. In the future, algorithm design can be further optimized in the direction of adaptive weight allocation and structured time modeling, and extended to deep dynamic cognitive analysis in cross domain collaborative learning and intelligent education scenarios.

AUTHOR CONTRIBUTIONS

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With an overall recall rate of 80.3% across all six forms of transfer, the hybrid fusion feature model maintains a high degree of accuracy with the greatest recall achieved while transferring between an organiser and an observer (84.3%) being indicative of how well the model captures change during decision-making processes and coordinated behaviours. The lowest recall rate occurred during transfers between an observer to an collaborators (77.9%), mostly due to the effect of the reverse transfer of role semantics. In general, the data shows that the multi-headed attention mechanism aids in capturing cross-temporal features associated with role transitions.

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